

SASWAT K. NAYAK

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EDUCATION

Master of Science(Machine Learning in 3D Photonics), University of Wisconsin-Madison Aug 2024

- Minor in **Computer Science**.

Bachelor Degree, Indian Institute of Technology Kharagpur June, 2021

RELEVANT COURSES

Artificial Neural Networks, Software Engineering, Image Processing, Advanced Machine Learning, Deep Learning, Information Retrieval, Fluid Mechanics, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Statistical Inference.

SKILLS

Technical Skills	Javascript, Python, CI/CD, AWS, Docker, Gitlab, Pandas,R, Matplotlib, SQL, Tableau, GCP, R, Hive, MongoDB, Snowflake, Databricks, AWS Lambda, HTC, Agile, Scrum Master, Azure, Inferential methods, A/B testing, Kafka, Athena, MEDDPIC
Machine Learning	NLP, Basic Computer Vision, Transformers, Tensorflow, PyTorch, GANs, Language Models. Pandas, Numpy, MLOps, Spacy, Large Language Models(BERT, GPT-2, T5, RoBERTa), Big Data Engineering, Elasticsearch, XGBoost, Random Forest, Scikit-learn, Langchain.

EXPERIENCE

Projects [Live Demo](#)

- Execute an end-to-end data engineering project on real-time stock market data using Kafka, Amazon Web Services (AWS), Apache Kafka, Glue, Athena, and SQL.
- Engineered a high-performance filesystem backed by relational databases, featuring a user-friendly web interface for comprehensive CRUD operations. Spearheaded the development of an intuitive visualization tool, enabling users to effortlessly explore directories and subdirectories through a dynamic tree structure. Deployed on Vercel.
- Used LangChain and Llama2 to make my own custom chatbot which answers questions based on my professional experience. Made it 70% faster using embedding, vector stores and optimized document chunking.

Data Analyst(Full Time) Innovaccer Analytics(Healthcare) Jul 2022

- Enhanced query efficiency by 40%, transitioning our knowledge base from MySQL to Neo4j. I advocated to the Vice President the benefits of Neo4j for our specific needs, highlighting its adeptness at managing complex, interconnected data, thereby streamlining our operations and setting the stage for scalable growth.
- Led a team of interns to extract patient data from progress notes, achieving a 90% reduction in time required and a 50% decrease in associated costs by implementing NLP solutions(NLTK, Spacy, RoBERTa) and using auto-encoders for anomaly detection.
- Developed and implemented RESTful APIs to improve healthcare data operations. Applied DevOps(Docker, Kubernetes) practices, boosting deployment efficiency by 30% and fostering team collaboration.

GRA: Machine Learning in 3D Photonics Sept 2022 - Present 2023 — *Madison, WI*

- Introduced novel explainable descriptors that significantly outperform traditional symmetry-based descriptors in machine learning tasks. These advancements achieved a notable 50% improvement in the effectiveness of identifying and categorizing photonic crystal structures, as substantiated by an F1-score enhancement from 48% to 94%.

Machine Learning Intern, Ushur Nov 2020 - Dec 2020

- Designed and implemented an advanced algorithm combining Google Tesseract OCR with rule-based techniques to accurately extract customer data from insurance claim documents; securing victory in an internal hackathon.